

Output
Each number of rows (half of diamond height) : 3

```

      *
    * * *
  * * * * *
    * * *
      *
  
```

Aim:- Write a shell programming to print different shapes - Diamond, Rectangle, Triangle, Hollow square, square.

1) Diamond:-

read -p "Enter number of rows (half of diamond height) : " n

for ((i=1; i<=n; i++))

do

for ((j=1; j<=n; j++))

do

echo -n " "

done

for ((j=1; j<=2*i-1; j++))

do

echo -n " * "

done

echo

done

for ((i=n-1; i>=1; i--))

do

for ((j=n; j>=1; j--))

do

echo -n " "

done

for ((j=1; j<=2*i-1; j--))

do

echo -n " * "

done

echo

done

output

Enter number of rows : 3

Enter number of columns : 5

```
* * * * *
* * * * *
* * * * *
```

output

Enter number of rows : 4

```
*
* *
* * *
* * * *
```

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(2) Rectangle :-

```
read -p "Enter number of rows : " rows
read -p "Enter number of columns : " cols
for ((i=1; i<=rows; i++))
do
for ((j=1; j<=cols; j++))
do
echo -n "x"
done
echo
done
```

(3) Triangle :-

```
read -p "Enter number of rows : " n
for ((i=1; i<=n; i++))
do
for ((j=1; j<=i; j++))
do
echo -n "x"
done
echo
done
```

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output

Enter size of Square: 5

```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```

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4) Hollow Square

```
read -p "Enter size of square : " n
for ((i=1; i<=n; i++))
do
    echo -n "*"
done
echo
for ((i=2; i<=n; i++))
do
    for echo -n "*"
for ((j=2; j<=n; j++))
do
    echo -n " "
done
echo " "
done
echo "x"
done
if ((n>1))
then
for ((i=1; i<=n; i++))
do
    echo -n "*"
done
echo
fi
```

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Output

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

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5) Square

Size = 5

for ((i=1; i<=Size; i++))

do

for ((j=1; j<=Size; j++))

do

echo -n "*"

done

echo

done

Viva Question

Q1 How do you loop from 1 to 5 in Bash for i in
1

Ans for i in {1..5}

do

echo \$i

done

Q2 What does \$i represent in a for loop?

Ans It holds the current value of the loop variable.

Q3 What is the syntax for loop?

Ans for (())

do

done

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Q4. What is the purpose of the keyword "in" for loop?
Ans. The "in" keyword in a for loop is used to specify the list or range of values that the loop will iterate over.

Q5. How to terminate for loop?
Ans. Using break keyword.

output

Enter ~~output~~ number of element : 5

Enter 5 elements:

1 2 3 4 5

Array elements:

1 2 3 4 5

output

Enter number of element : 5

Enter 5 elements.

10 20 30 40 50

Reversed array

50 40 30 20 10

Aim:- Shell programming - Arrays

(A) Write a program to read & print elements of array.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[100], n;
```

```
    printf("Enter number of elements : ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter %d element : \n", n);
```

```
    for (int i=0; i<n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    printf("Array element : \n");
```

```
    for (int i=0; i<n; i++)
```

```
        printf("%d", arr[i]);
```

```
    return 0;
```

```
}
```

(B) Write a C program to find sum of all array elements.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[100], n, sum=0;
```

```
    printf("Enter number of elements : ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter %d element : \n", n);
```

```
    for (int i=0; i<n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
        sum += arr[i];
```

```
}
```

Signature _____

output

Enter number of elements: 3

Enter 3 elements

4 7 2

Enter element to search: 2

Element found at index 2

```
printf("Sum of elements: %d", sum);  
return 0;  
}
```

Q) Write a C program to find ~~reverse~~ search on element in an array.

```
#include <stdio.h>  
int main() {  
    int arr[100], n, key, found = 0;  
    printf("Enter number of elements: ");  
    scanf("%d", &n);  
    printf("Enter %d elements: \n", n);  
    for(int i = 0; i < n; i++)  
        scanf("%d", &arr[i]);  
    for(int i = 0; i < n; i++) {  
        if(arr[i] == key) {  
            printf("Element found at index %d \n", i);  
            found = 1;  
            break; }  
    }  
    if(!found)  
        printf("Element not found. \n");  
    return 0;  
}
```

Output

Enter ~~output~~ number of elements : 4

Enter elements :

3 1 4 2

Sorted array :

1 2 3 4

5. Write a C program to sort array's elements in ascending or descending order.

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[100], n, i, j, temp;
```

```
    printf("Enter number of elements : ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements : \n");
```

```
    for (i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    for (i = 0; i < n - 1; i++) {
```

```
        for (j = i + 1; j < n; j++)
```

```
            if (a[i] > a[j]) {
```

```
                temp = a[i];
```

```
                a[i] = a[j];
```

```
                a[j] = temp;
```

```
            }
```

```
    printf("Sorted array : \n");
```

```
    for (i = 0; i < n; i++)
```

```
        printf("%d", a[i]);
```

```
    return 0;
```

```
}
```


Viva Question

Q1 What is an array in C?

Ans A array is a collection of elements of the same data type stored in contiguous memory location.

Q2 Why are array indexing?

Ans Array indexing refers to accessing elements using position numbers starting from index 0.

Q3 Why are array used in programming?

Ans Array are used to store and process multiple values using a single variable name.

Q4 Which loop is commonly used with arrays?

Ans Sorting arranges array elements in ascending or descending order. For loop is commonly used to read, display, search & sort.

Q5 What is sorting in an array?

Ans Sorting arranges array elements in ascending or descending order.